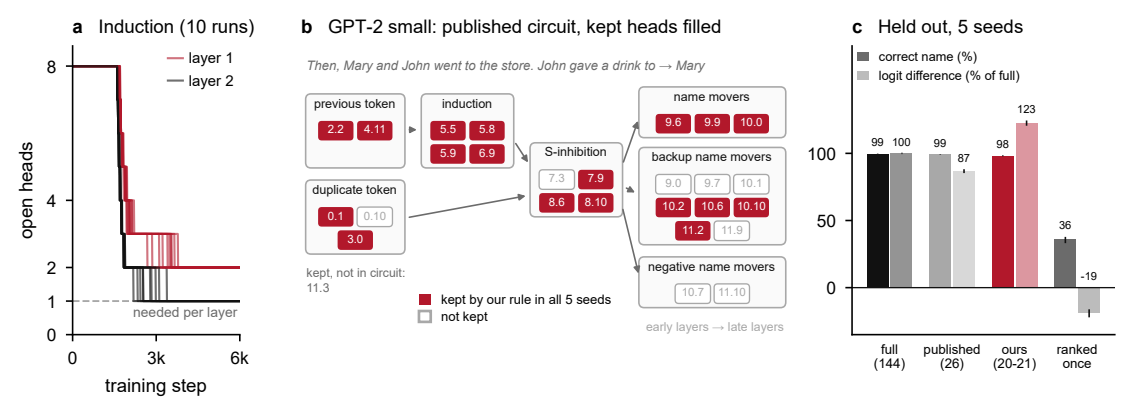

a
Induction (10 runs)
layer 1
layer 2
open heads
8
4
2
1
0
needed per layer
0
3k
6k
training step
b
GPT-2 small: published circuit, kept heads filled
Then, Mary and John went to the store. John gave a drink to → Mary
previous token
2.2
4.11
induction
5.5
5.8
5.9
6.9
duplicate token
0.1
0.10
3.0
S-inhibition
7.3
7.9
8.6
8.10
name movers
9.6
9.9
10.0
backup name movers
9.0
9.7
10.1
10.2
10.6
10.10
11.2
11.9
negative name movers
10.7
11.10
kept, not in circuit:
11.3
kept by our rule in all 5 seeds
not kept
early layers → late layers
c
Held out, 5 seeds
correct name (%)
logit difference (% of full)
100
50
0
99
100
99
87
98
123
36
-19
full
(144)
published
(26)
ours
(20-21)
ranked
once